The Downside of More: Choice Overload in Health Insurance Markets

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Abstract

Health insurance markets often offer multiple plans, yet consumers frequently choose strictly dominated options. Using two natural experiments in China's private health insurance market, we study how choice set size affects choice quality. An insurer exogenously expanded renewal menus from two to three plans and later reduced them back to two. Exploiting strictly dominated plans and a regression discontinuity design, we find that adding one option increases dominated choices by 8.4 percentage points (52%), while removing one reduces them by 11.0 points (49%). Evidence points to choice overload rather than changes in average plan quality as the main reason. *JEL Classification:* G22, I13. *Keywords*: Health insurance; Dominated plan; Choice set size; Choice overload.

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1. Introduction

Health insurance choice is important because it is a point of entry into decision-making in the health care sector (Chandra et al., 2019). In many markets, consumers must choose from health insurance menus with multiple options. However, the literature has documented that consumers often choose suboptimal plans or even dominated plans (Abaluck and Gruber, 2011, 2023; Bhargava et al., 2017; Handel, 2013; Handel et al., 2024; Liu and Sydnor, 2022; Samek and Sydnor, 2025).[5]

There remains a debate about ways to design effective policies to enhance consumers' decision quality in this context. For example, prior literature has found mixed evidence on whether consumers learn over time (Abaluck and Gruber, 2016a, 2016b; Ketcham et al., 2012, 2016) and on whether information interventions improve choice consistency (Bundorf et al., 2019; Ericson et al., 2017; Gruber et al., 2020; Kling et al., 2012; Samek and Sydnor, 2025). The literature has also documented that changing choice architecture, including standardizing plan options (Ericson and Starc, 2016) or reducing choice set size (Abaluck and Gruber, 2023), can enhance choice quality.

In this paper, we build on this literature and explore how choice set size affects consumers' choice quality, which has received limited scrutiny, as reported by Abaluck and Gruber (2023). We leverage a setting in which individuals purchase a private health insurance plan from multiple plan options. Our data are from a leading private health insurance company that exogenously changed the size of its menu from two to three plans and then from three to two plans in two consecutive natural experiments. Another important feature of our setting is the prevalence of strictly dominated plans within each menu. This allows us to derive an unambiguous and consistent measure of consumer choice quality across different menus without imposing additional assumptions on loss distributions and risk preferences.

We apply a regression discontinuity in time (RDiT) design to identify the causal impact of changing the choice set size on consumer choice quality. We provide evidence of identification validity by showing that there is no discontinuous

[5] Dominated plans generate higher total spending (plan premiums plus out-of-pocket medical spending) at any level of medical spending, compared to their alternatives (see Liu and Sydnor, 2022, for a detailed discussion).

change in the density of the running variable (McCrary, 2008) or in the values of the covariates at the cutoff dates.

We find that increasing the choice set size leads to a decline in consumer choice quality, and conversely, reducing the choice set size improves choice quality. Specifically, we show that adding one insurance plan to a two-plan menu significantly increases the probability of choosing a strictly dominated plan by around 8.4 percentage points in absolute terms, or 52% relative to the baseline mean before the menu change. In contrast, reducing the choice set size from three to two plans leads to a decrease of 11.0 percentage points in choosing dominated plans, or 49% in relative terms. The two estimates are statistically close to each other, suggesting the effects are likely to be symmetric.

To rule out unobserved contemporaneous shocks, we additionally conduct a placebo test using a separate sample of enrollees, whose plan menu remains unchanged at the cutoff date of the second natural experiment and includes a strictly dominated plan. For this group, we find no evidence of a discontinuous change in the probability of choosing the dominated plan at the same cutoff date.

We further investigate the potential mechanism behind the choice of dominated plans and how changing the choice set size has an impact on it. We focus on two main channels suggested by the recent literature: choice overload (e.g., Besedeš et al., 2015) and changes in the average plan quality associated with the menu change (e.g., Abaluck and Gruber, 2023). Our findings are more supportive of the choice overload mechanism.

We first explore whether providing more options leads to worse choice quality because of choice overload, that is, whether consumers' choice functions perform worse in the face of larger choice sets. Several stylized facts in our data support this mechanism. First, we find that more-experienced consumers make better choices before the menu changes and are less affected by the changes in choice set size. Second, we leverage another natural experiment at the firm, in which some insurance agents are trained with artificial intelligence (AI) tools to help their customers with plan selection. Although the program improves choice quality in general, the improvement is only noticeable among consumers choosing from three-plan menus, that is, the group featuring the lower baseline choice quality. This result suggests that information interventions can mitigate the performance gap in choice functions between the two-plan and three-plan menu

settings. Third, we find that consumers seem to be more satisfied with their chosen plans when selecting from a smaller menu.[6] This result is consistent with the previous literature on choice overload, which documents that larger choice sets can reduce satisfaction with the chosen options (e.g., Iyengar and Lepper, 2000; Scheibehenne et al., 2010).

Next, we explore whether providing more options leads to worse choices because the average plan quality declines as the plan menu grows larger. We measure the average plan quality using the fraction of plans that are dominated.[7] For more than half of the consumers, the proportion of dominated plans is 50% in a two-plan menu (one dominated plan) and 67% in a three-plan menu (two dominated plans). Thus, worse choice quality under a three-plan menu is consistent with the corresponding deterioration in average plan quality. However, the other consumers (around 40%) face only one dominated option in both the two-plan and three-plan menus (so the average quality is lower under the two-plan menu). We find that these consumers are still more likely to choose dominated options under the three-plan menu when the average plan quality improves, suggesting that variation in the average plan quality is not the driving force causing worse choice quality.

Our work builds on a large body of literature on choice frictions in health insurance markets. Our contributions are threefold. First, we add to a growing strand of literature exploring the impact of choice set size on health insurance choice quality (e.g., Abaluck and Gruber, 2023; Bhargava et al., 2017). The findings have been somewhat mixed. Some survey experiments have found no significant link between menu size and consumers choice quality when plans vary along fewer dimensions (e.g., Bhargava et al., 2017). However, recent field studies have found that consumer choices are improved under a reduced number of plans (e.g., Abaluck and Gruber, 2023). We connect these findings by demonstrating that plan menu reduction can improve choice quality even when the original

[6] Following the spirit of the existing literature (e.g., Gruber et al., 2020; Ketcham et al., 2012, 2015), our choice satisfaction measure is derived from the enrollee's decision to renew automatically their current chosen options for the following year. Gruber et al. (2020) and Ketcham et al. (2012, 2015) utilize plan switching as an ex-post measure of plan satisfaction. In contrast, our measure captures the complementary dimension: the intention to remain in the chosen plan.

[7] The measure of the average plan quality that we construct in the baseline analysis is more comprehensive than the description provided here, as it also accounts for the choice of riders (optional coverage for specific medical services) when determining plan dominance. Nonetheless, this adjusted choice quality measure yields similar insights.

choice set is small, provided the number of plan attributes for comparison is large. Within our setting, consumers choose from a small choice set of no more than three plans, yet they must evaluate around 20 plan characteristics—in both financial and nonfinancial terms.[8]

Second, our findings add to the discussion on choice overload in the context of health insurance (e.g., Abaluck and Gruber, 2023; Ketcham et al., 2015; Li et al., 2025). Our study differs from prior work by documenting choice overload as the primary driver of consumer choice quality. We also show that other explanations, such as changes in average plan quality (Abaluck and Gruber, 2023), are of secondary importance in our context.

Third, we contribute to the strand of literature on health insurance choice frictions in the developing countries, including China, supplementing existing studies based on experiences in developed countries. This extension is meaningful, as the previous literature find that health insurance choice patterns vary significantly with socioeconomic factors (Handel et al., 2024). We show that in developing countries like China, where consumers are not experienced in navigating health insurance plans (the majority of the population is covered by the basic health insurance program without making any active plan choices, and purchasing private health insurance is optional), choice inconsistency can happen even with a small choice set. Our findings also highlight that it is necessary for regulators and insurers to take care in designing health insurance menus in this context.

The rest of the paper is organized as follows. Section 2 provides a brief introduction to the private health insurance plans we sample, as well as details of the natural experiments. Section 3 presents our data. Section 4 describes the identification strategy and discusses identification validity. Section 5 presents the main results and robustness checks. Section 6 discusses the potential mechanisms. Section 7 concludes with policy implications.

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[8] On the one hand, prior studies consider a larger number of available plans than we do. For example, the number of plans varies from four to twelve in Bhargava et al. (2017). Abaluck and Gruber (2023) study menus containing two to seven plans, and the average number of plans in their main sample is around four. In contrast, the size of our choice set varies from two to three. On the other hand, prior studies consider fewer plan attributes than we do. For example, in Bhargava et al. (2017), the number of plan attributes vary from two (the deductible and maximum out-of-pocket spending) to one (the deductible). Abaluck and Gruber (2023) hold all nonfinancial characteristics of plans constant. In comparison, within our setting, consumers compare a dozen plan characteristics—both financial (e.g., the deductible and the premium) and nonfinancial (e.g., items covered for treatment).

2. The health insurance plans and natural experiments

2.1 The private health insurance market in China and the sampled product

In China, the public basic medical insurance program mainly targets essential health needs and provides a foundation of protection, offering very limited choices. Most employed residents are assigned a single plan, while others typically face only a couple options (typically no more than two plans). Private health insurance plans, which serve as an important supplement to the public basic medical insurance program (Liu et al., 2011), typically offer a range of plan options for individuals to choose and purchase on a voluntary basis.

Because short-term insurance products of one-year duration dominate China's private health insurance market, policyholders must renew their plans annually to obtain long-term coverage. At renewal, an insured individual is granted the right to make an active choice among the different insurance plans offered by the insurer.

Consumers in China's private health insurance market are predominantly young (approximately 55% are under age 35) and highly educated (with over 65% holding a college degree), as reported by Zhu et al. (2024). However, these consumers do not have experience in navigating health insurance plans because there are few variations under the public basic medical insurance. This distinguishes them from the consumers in developed markets, which have been studied extensively, such as Medicare Part D beneficiaries. The documented links between socioeconomic factors and health insurance choice patterns (e.g., Handel et al., 2024) underscore the need to assess the external validity of previous findings on choice frictions in diverse contexts. China's private health insurance market provides a unique laboratory to examine whether choice frictions persist and how they vary with the size of the choice set among young, highly educated, but less experienced market participants.

Our empirical analysis is based on a data set acquired from a leading private health insurance provider in China. The provider offered a popular insurance product for inpatient care expenses.[9] First launched in 2016, the product was a great success in the market and became mainstream in China's private health insurance market.

[9] The insurance product covers inpatient medical expenses and expenses associated with treatment of critical illness that exceed the coverage cap for public basic medical insurance.

2.2 The natural experiments

We study the effect of changing the choice set size on insurance choice quality. Our empirical analysis leverages two natural experiments that altered the renewal menus for existing enrollees. These menu changes were exogenous for the following reasons. The menu changes stemmed from supply-side adjustments and were orthogonal to localized or individual-level demand shocks. The changes took place without any preannouncement and took effect for all affected enrollees at the same time; thus, the changes should not have been anticipated by the consumers. As policyholders could only renew through personalized renewal links that directed them to predetermined renewal menus, consumers' strategic manipulation of the choice sets was also unlikely.

The choice menus in our analysis involve four plans. For ease of expression, we refer to the standard plans launched at the start of 2019, 2020, and 2021 as "Plan A," "Plan B," and "Plan D," respectively. The family plan launched in mid-2020 is denoted "Plan C."

The timeline of the natural experiments and detailed choice menus during our observation period (2020–21) are summarized in Figure 1: (i) Menu 1: before July 2020, individuals faced two renewal options—their original choice from the previous year (Plan A) and the upgraded Plan B; (ii) Menu 2: since July 1, 2020, Plan C was included in the menu, and the other two plans remained the same; (iii) Menu 3: since January 29, 2021, Plan B and Plan C were replaced by an integrated plan (Plan D), bringing it down to a two-plan menu—Plan A and Plan D. The transitions from Menu 1 to Menu 2 and from Menu 2 to Menu 3 resulted in two exogenous variations in the size of the plan menu for the enrollees, allowing us to identify the causal effect of changing the choice set size on consumer choice quality.

The choice architecture involved a small menu but high information density. On the online renewal choice interface, consumers were presented with around 20 plan characteristics—both financial terms (e.g., premium, premium discount, and deductible) and nonfinancial terms (e.g., covered treatment items).

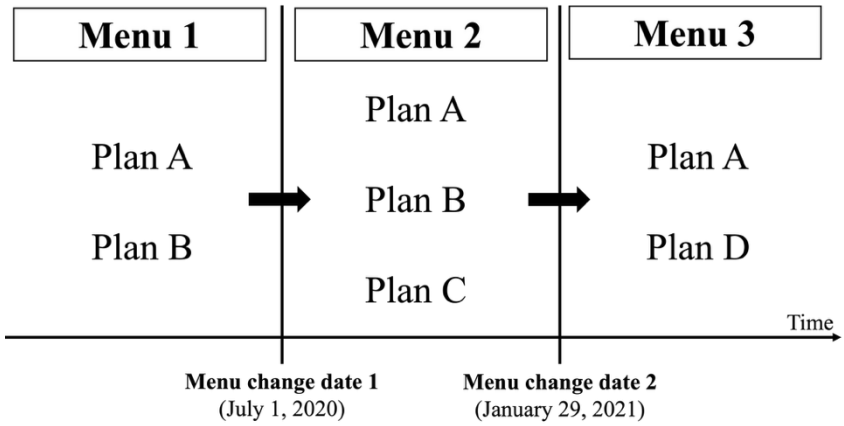


Figure. 1: Timeline of natural experiments and plan menu changes

3. Data and sample

3.1 Sample construction

Our empirical analysis is based on a dataset acquired from the private health insurance product provider. Our raw dataset consists of individuals who were originally enrolled in 2017 or 2018 (the insurance product was first introduced to the market at the end of 2016). We restrict the sample to individuals maintaining continuous coverage from initial enrollment through the years of the natural experiments, who held Plan A in the years preceding each intervention. For these individuals, we have information on their choice sets (as shown in Figure 1) and plan choices throughout both natural experiments.

To obtain cleaner estimates, we impose two additional sample restrictions. First, we exclude individuals who placed their renewal orders on the exact dates of the menu changes. Because these menu changes took place in the middle of the day (and we only observe daily-level data), removing observations from the policy change dates avoids a downward bias in the estimated treatment effects.[10] Second, to ensure comparability, we only keep those who placed renewal orders within a 60-day bandwidth before and after each cutoff date.

[10] In the following analysis, we define the dates immediately following the dates of menu changes as the cutoff dates (i.e., July 2, 2020, and January 30, 2021). These dates represent the first renewal dates when all remaining individuals were subject to the treatment and assigned to the new menus.

3.2 Consumer choice quality measure and additional sample restrictions

Our unique setting enables us to construct a consistent measure of consumer choice quality across different plan menus, defined by whether an individual chooses a strictly dominated plan (Bhargava et al., 2017; Liu and Sydnor, 2022). For an enrollee with any level of medical spending, a strictly dominated plan induces strictly higher total costs (out-of-pocket spending plus insurance premiums) compared to the strictly dominating options.

Our measure of choice quality has several advantages compared to other measures based on models of expected out-of-pocket spending (e.g., Abaluck and Gruber, 2023). First, our measure is free of assumptions about out-of-pocket spending, such as the perfect back-cast assumption (enrollees believe that next year's claims will be identical to those in the current year) or the perfect foresight assumption (enrollees know exactly what their out-of-pocket costs will be in the coming year). Moreover, our measure does not require individual-level data on medical spending—information that is not observed in our data.

Some plans offer riders that provide supplementary coverage in specific medical catalogs and services (e.g., executive VIP medical services) on the condition of purchasing the main plan. To determine plan dominance, we account for the choice of riders. A plan-by-rider combination strictly dominates another if it (i) offers the same or better insurance protection (e.g., a broader network) and (ii) charges an equal or lower premium, with at least one of these two conditions being strictly favorable to the consumer.

To ensure the existence of strictly dominated options within each menu, we impose two additional sample restrictions. We restrict the sample to enrollees with no claim history, and we exclude individuals age between 16 and 20 or over 60. These data restrictions ensure that all the individuals in our sample have a strictly dominated choice under all three menus. Appendix A.3 provides a more detailed discussion of the plan-by-rider options.

3.3 Summary statistics

Our final study sample consists of 37,888 and 29,614 individuals who placed their renewal orders within a 60-day bandwidth around the cutoff dates in the first and second natural experiments, respectively. Because the two menu changes took place within a seven-month period, and the plan was renewed on a yearly basis,

there is no overlapping of affected enrollees; thus, each observation represents a unique enrollee.

Table 1 presents summary statistics for the main covariates by natural experiment and choice menu. Significant differences emerge across several observed characteristics for individuals who made their choices under the different menus. For example, enrollees who chose before the menu change were slightly older than those who chose after the change in the first natural experiment (panel A). This indicates that a direct comparison between individuals below and above the cutoffs may lead to biased estimates due to confounders.

To address this issue, we employ a RDiT design as our identification strategy and restrict the comparison to a narrower bandwidth around the cutoff dates, following Calonico et al. (2014). In addition, we impose a triangular kernel to assign lower weights to observations further from the cutoffs in the main regressions. Section 4 provides a detailed discussion of the identification design and its validity. Table 2 demonstrates that we achieve covariate balance after further adjustments.

Table 1: Summary statistics of the main covariates

	(1) Before menu change	(2) After menu change	(3) Difference (after – before)
Panel A. Natural experiment 1: Increasing choice set size from two to three			
Age	37.540	37.239	-0.301*
Male	0.484	0.475	-0.010*
Household size (#)	1.691	1.755	0.064***
Any rider (previous year)	0.594	0.604	0.009*
Historical policies (#)	2.536	2.724	0.189***
Initial enrollment in 2017	0.532	0.719	0.187***
Obs.	24,211	13,677	-
Panel B. Natural experiment 2: Reducing choice set size from three to two			
Age	37.750	38.164	0.414**
Male	0.480	0.478	-0.002
Household size	1.903	1.854	-0.049***
Any rider (previous year)	0.659	0.591	-0.067***
Historical policies (#)	3.023	3.504	0.481***
Initial enrollment in 2017	0.435	0.500	0.065***
Obs.	13,434	16,180	-

Notes: This table reports the means of the baseline covariates by natural experiment and choice menu. "Historical policies (#)" is the total number of policies held before renewal in the natural experiments. "Initial enrollment in 2017" is a dummy variable that equals 1 for policyholders who initially enrolled in 2017 and 0 for those who enrolled in 2018.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4. Empirical design

4.1 Main identification strategy

We employ a RDiT design to estimate the effect of changing the menu size on consumer choice quality. Two natural experiments, as discussed in section 2.2, create exogenous variations in the choice sets. These variations took effect on July

2, 2020 (cutoff date 1) and January 30, 2021 (cutoff date 2).[11] We compare consumers' choices immediately prior to and following the menu changes. Specifically, we estimate the following equation:

$$Y_{it} = \alpha + \beta D_{it} + f(X_{it}) + \varepsilon_{it}, \quad (1)$$

where i and t index policyholders and time, respectively. $Y_{i,t}$ is the outcome variable of interest, indicating whether an individual selects a strictly dominated plan at renewal. D_{it} is a dummy variable indicating whether consumer i makes the plan choice under the new menu (Menu 2 and Menu 3 in the first and second natural experiments, respectively). X_{it} is the running variable representing the plan choice date (renewal date) relative to the cutoff date.

In our baseline regressions, we utilize a linear spline and allow for different trends across the threshold: $f(X_{it}) = \gamma_1 X_{it} + \gamma_2 D_{it} X_{it}$. We specify local linear regressions with a triangular kernel, which assigns lower weights to observations that are farther from the cutoffs. Standard errors are clustered at the calendar day level to account for potential correlation in the errors among enrollees who make plan choices on the same day.

For bandwidth selection, we follow the approach proposed by Calonico et al. (2014) for each regression and calculate that the optimal bandwidth is around 14 days. To enable comparisons across regressions, we follow the literature (e.g., Ang et al., 2024; Giuntella and Mazzonna, 2019; Lee and Vabson, 2024) and set a uniform bandwidth of 14 days across all outcomes in the main regression. The resulting effective numbers of observations in the baseline regressions are 7,699 and 5,433 for the first and second natural experiments, respectively. By doing so, we limit the analysis to a narrow time window, leveraging the large number of observations around the cutoffs. Our approach departs from empirical designs based on interrupted time-series data (e.g., daily observations for a given geographic region), which typically rely on observations distant from the cutoff dates and are thus more susceptible to confounders like other policy changes (see Hausman and Rapson, 2018, for a review).

[11] July 2, 2020, and January 30, 2021, are the days immediately following the menu changes. We exclude the exact dates of the menu changes (July 1, 2020, and January 29, 2021) to avoid downward bias in the estimated treatment effects, because they are partial treatment dates (the menu changes took place in the middle of the days), as mentioned in section 3.1.

4.2 Validity tests

Our key identification assumption is that individuals are unable to manipulate precisely whether they fall on one side of the discontinuity or the other. Since both menu changes took place without any announcement and individuals could only renew through personalized web links, such sorting is unlikely to be a problem in our setting. To provide empirical evidence, we test for discontinuity in the density of the running variable (McCrary, 2008) and find no evidence of excess bunching at either cutoff.[12]

We also check for imbalances at the cutoffs in terms of observable characteristics. Table 2 presents the results. As expected, we find no statistically significant differences across either cutoff in observed characteristics, including age, gender, household size, the previous year's rider choice, the total number of historical policies before renewal, and the initial enrollment year. Graphical evidence in Appendix Figures B.2 and B.3 also reveals no obvious discontinuities at either cutoff.

[12] The estimated discontinuities (standard errors) are 0.001 (0.047) and 0.075 (0.046) at the cutoffs for the first and second natural experiments (increasing and reducing the choice set sizes), respectively. See Appendix B.1 for details.

Table 2: Covariate balance

Variable	(1) Baseline mean	(2) RDiT estimate
Panel A. Natural experiment 1: Increasing choice set size from two to three		
Age	37.801	0.856
		(0.550)
Male	0.490	-0.011
		(0.013)
Household size (#)	1.680	-0.024
		(0.083)
Any rider previous year	0.598	-0.001
		(0.030)
Historical policies (#)	2.598	0.036
		(0.027)
Initial enrollment in 2017	0.598	0.034
		(0.027)
Eff. number of obs.	7,699	
Panel B. Natural experiment 2: Reducing choice set size from three to two		
Age	37.733	0.274
		(0.892)
Male	0.481	-0.027
		(0.022)
Household size (#)	1.960	-0.143
		(0.095)
Any rider previous year	0.629	-0.042
		(0.034)
Historical policies (#)	3.447	0.058
		(0.055)
Initial enrollment in 2017	0.468	0.070
		(0.046)
Eff. number of obs.	5,433	

Notes: "Baseline mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff. "RDiT estimate" denotes the baseline RDiT estimates.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5. Main results and robustness

5.1 Main results

This section presents the main results on the effect of changing choice set size on consumer choice quality. Figure 2 documents a sharp discontinuity in consumer choice quality on the days when the choice set sizes changed. As panel A shows, the probability of making choice mistakes, measured by the probability of choosing a strictly dominated plan at renewal, increases sharply after the menu size increases from two plans (Menu 1) to three plans (Menu 2). Consistent with our expectation, panel B shows that reducing the menu size from three plans (Menu 2) to two plans (Menu 3) results in improved consumer choice quality. The probability of choosing a strictly dominated plan decreases steeply at the cutoff date corresponding to the reduction in plan menu size.

Table 3 reports the point estimates. As shown in column (1), expanding the menu from two to three plans increases the probability of choosing a dominated plan by 8.4 percentage points. The estimated effect is highly significant at the 1% level and large in relative terms, at around 52% relative to the baseline mean of 16.1 percentage points. Mirroring this, we estimate that reducing the choice set size from a menu of three plans to a menu with two plans lowers the probability of choosing a dominated plan by 11.0 percentage points, or approximately 49% compared to the baseline mean (column (2)).

In sum, the findings from the two natural experiments yield a consistent pattern in how the choice set size impacts consumer choice quality, and the estimated sizes of the effect are significant and generally comparable.

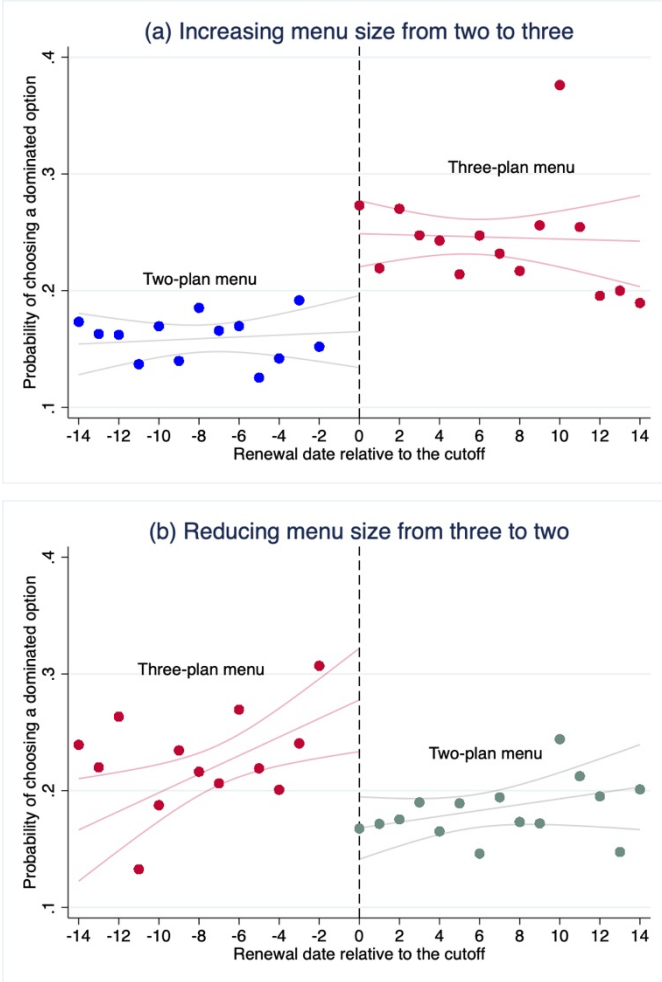


Figure 2: Change in menu size and consumer choice quality

Notes: Each panel plots the probability of choosing a strictly dominated plan at renewal for each standardized renewal-date bin, together with fitted lines and 95% confidence intervals. The fitted lines are produced using regressions with a triangular kernel. The first, second, and third choice menus are plotted in blue, red, and green, respectively.

Table 3: Effect of changing choice set size on consumer choice quality

	Outcome: Choosing a strictly dominated plan	
	(1)	(2)
	Increasing menu size from two to three	Reducing menu size from three to two
RDiT estimate	0.084***	-0.110***
	(0.023)	(0.031)
Eff. number of obs.	7,699	5,433
Below-cutoff mean	0.161	0.225

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

As a complementary analysis, we test whether the insurance renewal rate changes significantly in response to the menu changes. We employ an auxiliary data set from the same data source containing the original Plan A policies purchased a year before the natural experiments. These policies were scheduled to be open for renewal around the cutoff dates. We implement the same sample restrictions as in our baseline analysis and use this dataset to test whether the renewal rate changes significantly at the cutoffs. Appendix C reports the estimates, which show insignificant and close-to-zero changes in renewal rates around both cutoffs, suggesting that there are minimal extensive margin responses to the menu changes. This finding is noteworthy in its own right; furthermore, it confirms that our baseline findings are not driven by extensive-margin selection effects in the population of consumers who choose to renew at the cutoffs.

5.2 Specification and placebo tests

This subsection examines the sensitivity of the baseline estimates to alternative specifications and provides evidence to rule out unobserved contemporaneous shocks through a placebo test.

For specification checks, Appendix Table D.1 reports the results of implementing a donut-hole RDiT (excluding the nearest observations within a one-day radius around the cutoff), using a quadratic polynomial, utilizing a uniform kernel, employing the exact optimal bandwidth suggested by Calonico et al. (2014) for each regression, and adding covariates. The resulting point estimates remain highly consistent across different specifications.

Next, to rule out unobserved contemporaneous shocks, we conduct a placebo test using a separate sample of new enrollees of the same health insurance product. For these consumers, the choice menu does not change around the cutoff date of the second natural experiment (January 30, 2021). Specifically, since January 6, 2021, they choose from a stable menu consisting of two plans (Plan B and Plan D in Figure 1). For a subset of these individuals, Plan D strictly dominates Plan B, offering more generous coverage at the same premium. Because the plan menu remains unchanged at the cutoff, this setting provides a clean placebo test for detecting any concurrent shocks unrelated to the menu change in the second natural experiment.

We implement the placebo test by replicating the baseline RDiT analysis on this sample. To ensure the existence of strictly dominated plans, we impose sample restrictions consistent with those in the baseline analysis. We then test for any discontinuity at the cutoff in the probability of choosing a dominated plan.

The results are reported in Table 4. We find no evidence of a discontinuous change in choice quality at the cutoff in this placebo test. The estimated effect is small in magnitude (0.6 percentage points) and statistically insignificant. Consistent with this finding, the graphical analysis in Figure 3 shows no visible discontinuity at the cutoff. Appendix Table D.2 further shows consistent results from alternative RDiT specifications.

Table 4: Placebo test based on new enrollees

	Outcome: Choosing a strictly dominated plan
	(1)
RDiT estimate	0.006
	(0.013)
Eff. number of obs.	10,362
Below-cutoff mean	0.071

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who enrolled within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

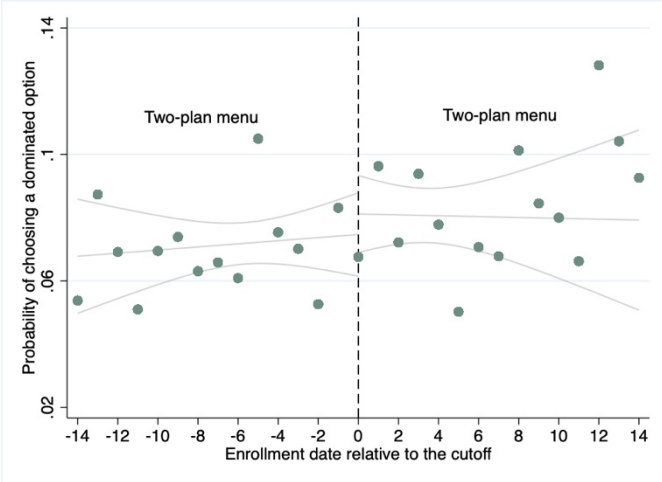


Figure 3: Placebo test based on new enrollees

Notes: Each panel plots the probability of choosing a strictly dominated plan for each standardized enrollment-date bin, together with fitted lines and 95% confidence intervals. The fitted lines are produced using regressions with a triangular kernel.

6. Discussion of mechanisms

The results reported in section 5 demonstrate that expanding choice set size lowers consumer choice quality, and reducing choice set size improves consumer choice quality. This section explores the potential mechanisms leading to these patterns. Recent literature has discussed two mechanisms: (i) changes in choice quality as a result of choice overload (e.g., Besedeš et al., 2015) and (ii) changes in average plan quality (e.g., Abaluck and Gruber, 2023). We examine the validity of these two possible mechanisms in our setting.

6.1 Choice overload

6.1.1 Consumers' enrollment experience

An alternative explanation of "choice overload" is that more options may translate into worse choices because consumers may feel overwhelmed when facing a large choice set (e.g., Besedeš et al., 2015). If this serves as an important mechanism, it would be expected that the negative treatment effect of expanding menu size would be smaller for individuals with better performance on baseline choice. The underlying assumption is that these individuals are more likely to choose wisely, even if their cognitive capacity is somewhat restricted by the larger menu, compared to those with worse performance on plan choice in the baseline period.

In our setting, individuals with better performance on baseline choice before the menu changes tend to have more years of insurance enrollment experience. This is consistent with consumer learning in health insurance markets (Ketcham et al., 2012). We split the estimation sample into "more experienced" and "less experienced" consumers based on the median number of historical enrollment records held by individuals at the time of renewal in the focal year. The more experienced group contains individuals with at least three (four) enrollment records, when the first (second) natural experiment took place. The less experienced group consists of consumers with fewer historical enrollment records than the sample median. The proportion of consumers making choice mistakes for the more experienced group is 0.151 (0.200) in the first (second) experiment before the menu change. As expected, the corresponding proportions are higher among the less experienced customers (0.176 and 0.246 in the first and second experiments before the menu changes, respectively).[13]

We then investigate the heterogeneous treatment effect across consumer enrollment experiences, to provide further evidence for choice overload. Figure 4 presents the RDiT estimates for each natural experiment by consumers' insurance experience. Appendix Table E.1 and Figure E.1, respectively, report the detailed estimates and figures. Consistent with the choice overload mechanism, the results show clearly heterogeneous results by consumer experience in both natural experiments. By and large, the group of more-experienced customers is not

[13] As a robustness check, we use an alternative definition of "more experienced" and "less experienced" consumers, based on the initial enrollment year. Specifically, we assign consumers who initially enrolled in 2017 to the "more experienced" group and those who enrolled in 2018 to the "less experienced" group. The results remain qualitatively similar to those reported and are available upon request.

affected by the change in the menu size, whether it increases or decreases. The estimated coefficients are insignificant and close to zero (0.030 and -0.028, respectively). In contrast, the less experienced customers are affected by the change in the size of the choice set in a consistent manner as shown in the main regression, that is, individuals make more choice mistakes when choose from a larger menu and fewer mistakes choosing from a smaller menu. The magnitudes of the estimates for the less-experienced group are substantially larger and highly significant for both natural experiments (0.141 and -0.171).

In sum, we find that changing choice set size has a large and significant impact on individuals with less enrollment experience, who also have a higher baseline probability of making choice mistakes. In contrast, the menu size change (in either direction) does not affect the choice quality of the more-experienced consumers. These findings indicate that consumers do not choose solely based on heuristics (e.g., Abaluck and Gruber, 2023) but can learn from their own enrollment experience (e.g., Ketcham et al., 2012), providing evidence supporting the existence of choice overload.

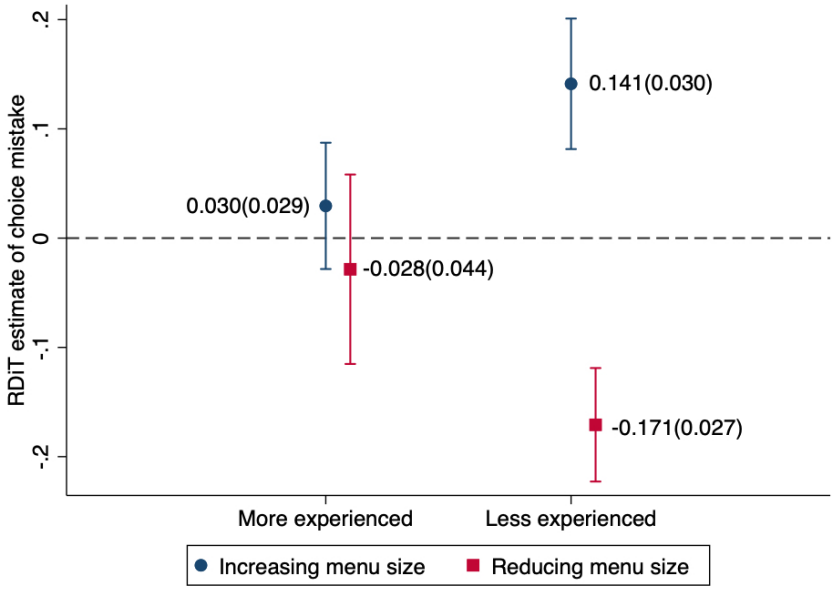


Figure 4: Heterogeneous treatment effects by baseline enrollment experience

Notes: This figure reports the RDiT estimates of choice mistake and 95% confidence intervals by natural experiment and consumer experience. The "more experienced" group contains individuals whose record of insurance enrollment history is equal to or above the sample median. The "less experienced" group contains those with fewer years of enrollment.

6.1.2 Decision support program

Another piece of evidence for the choice overload mechanism is provided by exploring the effect of a decision support program and how it varies across menu size. Our underlying assumption is that if choice overload is a mechanism contributing to enrollees' choice quality when the menu size changes, a decision support program should have a more pronounced effect within a larger menu, where individual choice functions are more likely to be weakened.

Aligning with the literature documenting that combining information provision with skilled agents may lead to choice quality improvements (Gruber et al., 2020), in November 2020, the insurer agent platform launched a decision support program aimed at improving consumer decision-making. It provided eligible agents with complimentary training sessions and AI-based tools. The training sessions offered courses taught by professors from leading universities on how to identify better plans for clients. An AI-based decision support tool was provided to agents in the program to help them analyze clients' needs and identify the best plans.

The timing of enrollment in this decision support program varies by insurance agent. For each sampled policyholder, we observe the calendar month when their agent first joined the program. Treatment turned on for those who renewed after the agent was enrolled in the program.[14]

Leveraging the staggered implementation of the decision support program, we employ a difference-in-differences (DID) design to investigate the impact of agents' decision support on consumers' choice quality. More importantly, we focus on how this effect varies across menu size.[15]

We impose the following sample restrictions to obtain the estimation sample. First, we apply the same sample selection criteria as in section 3 to ensure the existence of strictly dominated options. Second, to isolate the effect of the decision support

[14] For the sampled individuals, client-agent matching occurred during their initial enrollment in 2017 and 2018, years prior to the implementation of the decision support program.

[15] Alternatively, one could test how the treatment effect of changing choice set size varies between individuals with and without decision support in natural experiment 2. However, comparing RDiT estimates across these subsamples is precluded by a lack of observations near the cutoff. In the main estimation sample—those who renewed within 14 days around the cutoff—fewer than 50 individuals were affected by the program, rendering RDiT estimates for this group statistically unreliable.

program and avoid confounding influences from the menu change, we exclude individuals who renewed under Menu 1, leaving those who renewed under Menu 2 (from July 10, 2020, to January 25, 2021) and Menu 3 (from February 1 to March 31, 2021). Third, we exclude observations with missing values for the date when the agent signed up for the decision support program.

Following the literature on DID estimators with multiple treatments (changing the plan menu and providing decision support) and staggered treatment timings (e.g., Callaway et al., 2024; de Chaisemartin and d'Haultfoeuille，2023, 2024), we estimate the treatment effects of the decision support program on choice quality separately for different menus. We use the estimator of Callaway and Sant'Anna (2021) to address treatment-effect heterogeneity. As recommended, we only use the never-treated individuals as the control group.[16] Standard errors are clustered at the calendar date level as in the baseline RDiT specification. To show robustness, we also use the not-yet-treated individuals as the control group, and the results are consistent.

Moreover, we estimate the following two-way fixed effects estimator:

$$Y_{igt} = \alpha_0 + \beta Treat_g \times Post_t + \lambda_g + \Gamma_t + \varepsilon_{igt}, \qquad (2)$$

where i, g, and t index policyholders, treatment groups, and time, respectively. Y_{igt} is the measure of choice mistakes, an indicator of whether an individual chooses a strictly dominated plan at renewal. $Treat_g$ is the treatment group indicator, which equals 1 only for those individuals whose agents joined the support program within the observation period of each regression sample. $Post_t$ is the post-treatment period indicator, which equals 1 only for those who renewed during or after the months when their agents joined the program. λ_g denotes initial treatment month fixed effects, representing the calendar month when the agent first joined the program. The omitted reference group is the control group (never treated). Γ_t denotes calendar month fixed effects.

[16] The "never-treated" group is defined as individuals whose associated agents had not joined the program by December 2021, the final month for which data are available. The "not-yet-treated" group consists of those whose agents joined the program after the final month of the respective regression sample but no later than December 2021. For the sample choosing under Menu 2, the baseline estimation includes 175 treated, 1,911 not-yet-treated, and 46,871 never-treated observations. Among individuals choosing under Menu 3, there are 175 treated, 690 not-yet-treated, and 14,904 never-treated observations, respectively.

Table 5 reports the estimates of the impact of decision support on choice quality by choice menu. As shown in column (1), the decision support program significantly lowers the probability of customers choice mistakes by 22.1 percentage points under the three-plan menu (Menu 2). In contrast, the treatment effect is relatively small in magnitude and insignificant for the group choosing from Menu 3 (which contains two plans), with an estimate of 3.8 percentage points in column (2). Appendix Table E.2 confirms the robustness of these results by using the not-yet-treated group as the control, as well as employing a two-way fixed effects estimator. The results are highly comparable across methods.

Although the point estimate for the decision support program (-0.221 in Table 5) is larger in magnitude than that of reducing the choice set size (-0.110 in Table 3), their 95% confidence intervals overlap ([-0.434, -0.007] and [-0.170, -0.050], respectively). The effect of the decision support program is estimated less precisely because of the relatively smaller number of observations in the treatment group (fewer than 200 treated enrollees among 47,046 observations, shown in column (1) in Table 5).

Table 5: Effect of decision support on choice quality by plan menu

	Outcome: Choosing a strictly dominated option	
	(1)	(2)
	Choosing under Menu 2 (three-plan menu)	Choosing under Menu 3 (two-plan menu)
DID estimate	-0.221**	0.038
	(0.109)	(0.062)
Obs.	47,046	15,079
Pre-treatment mean	0.294	0.112

Notes: "Pre-treatment mean" reports the unconditional sample mean of the outcome for the treatment group policyholders during the pre-treatment period. The estimation sample here departs from that in the baseline RDiT analysis by: (i) restricting the scope to individuals choosing under Menu 2 and Menu 3; (ii) excluding observations with missing data on agents' enrollment in the decision-aid program; and (iii) extending the sampling window to maximize the inclusion of ever-treated individuals.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Figure 5 presents the corresponding event study estimates. The figure provides evidence supporting the validity of our DID strategy by showing parallel pre-trends. It also shows a sharp drop in the probability of choice mistakes immediately following treatment for those making choices under Menu 2 (three-

plan menu), while showing a relatively negligible change for the group choosing under Menu 3 (two-plan menu).

In sum, the findings indicate that agents' participation in a decision support program is effective in reducing their customers' choice mistakes when the menu is larger, where the baseline probability of making choice mistakes is high at 0.294. In contrast, when a smaller menu with two plans is provided, the probability of making choice mistakes at baseline is much lower (0.112). Therefore, providing a decision support program seems irrelevant to the outcome measure of choice quality. Taken together, these findings provide additional suggestive evidence supporting the choice overload mechanism.

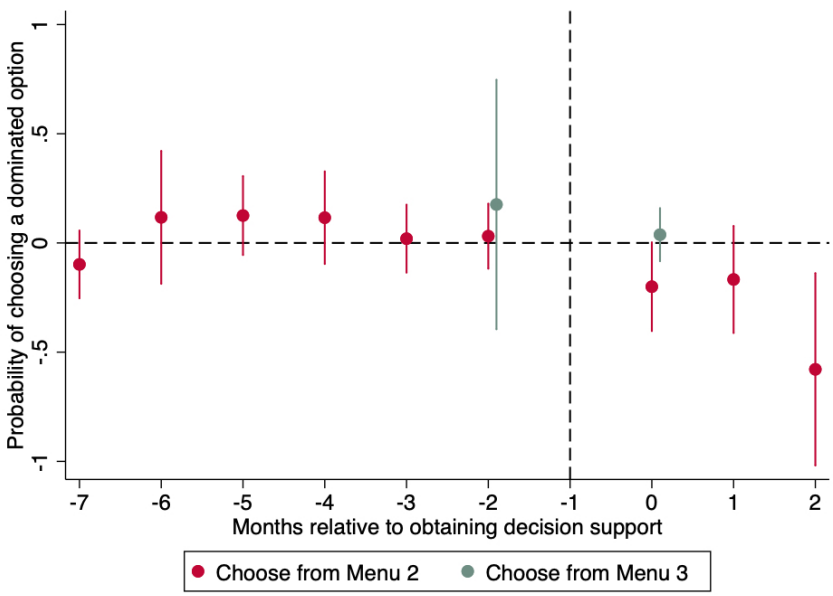


Figure 5: Event study estimates of the impact of the decision support program on the probability of making choice mistakes

Notes: The figure shows 95% confidence intervals.

6.1.3 Consumers' satisfaction with the chosen option

Previous literature on "choice overload" has documented that consumers may be more satisfied with their chosen options when choosing from a smaller choice set (Iyengar and Lepper, 2000; Scheibehenne et al., 2010). We test this prediction by utilizing additional information collected in the second natural experiment (reducing choice set size).

Following the literature, we measure consumer satisfaction with the chosen plans based on the stated preference to stay with the chosen option in the following year.[17] In the second natural experiment, immediately after the customer chose a plan and made the payment, the insurer sought customer authorization for automatic renewal of their chosen option in the future. The insurer asked customers whether they would agree to automatic renewal of the currently chosen options in the next year, with the payment deducted from the same account. The answers were binary ("yes" or "no"), and individuals were allowed to modify their choice throughout the entire coverage year. Our data capture each enrollee's final answer to this question recorded at the end of the coverage year, thereby incorporating any changes made during that period.

As shown in Table 6, reducing the plan menu size increases the probability that the customer signs up for automatic renewal by 4.4 percentage points. The estimated effect is significant and large, representing an increase of approximately 36% relative to the baseline mean of 12.3 percentage points. Appendix Figure E.2 confirms the sharp increase in this consumer satisfaction measure at the menu change cutoff. This finding is also consistent with the literature on choice overload, which links smaller choice sets to higher consumer satisfaction (Iyengar and Lepper, 2000; Scheibehenne et al., 2010).

Table 6: Effect of reducing the size of the choice set on consumer satisfaction with the chosen option

	Outcome: Automatic renewal
	(1)
	Reducing menu size from three to two
RDiT estimate	0.044*
	(0.025)
Eff. number of obs.	5,433
Below-cutoff mean	0.123

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[17] Gruber et al. (2020) and Ketcham et al. (2012, 2015) utilized plan switching as an ex-post measure of plan satisfaction. Our approach aligns with this literature by focusing on the intention to remain in the chosen plan.

6.2 Changes in average plan quality

Abaluck and Gruber (2023) document that facing more options leads to worse choices, mainly because the average plan quality declines as the plan menu expands. This is further explained by the fact that menu managers tend to prioritize better insurance plans in smaller menus.

We present key evidence showing that changes in average plan quality alone are not sufficient to explain our main findings on the impact of changing choice set sizes on consumer choice quality.

We first examine how menu changes affect average plan quality and then verify whether this effect aligns with the observed changes in consumer choice quality. To do so, we utilize the baseline RDiT strategy and replace the outcome variable with our measure of average plan quality, which is the proportion of strictly dominated options within a menu. This variable is defined as the ratio of the number of strictly dominated options (see Appendix Table A.2) to the total number of available combinations within a menu (see Appendix Table A.1).

Table 7 presents the RDiT estimates for average plan quality in both natural experiments. At first glance, the estimates are, in general, consistent with those of Abaluck and Gruber (2023). The estimates in columns (1) and (2) (0.007 and -0.053, respectively) show that the proportion of dominated options within the menu increases significantly in the first experiment (expanding menu size) and decreases in the second experiment (reducing menu size). The directions of these estimated effects mirror the baseline estimates for consumer choice mistakes, which are 0.084 and -0.110 for the first and second natural experiments, respectively (Table 3).

Table 7: Effect of changing choice set size on average plan quality

	Outcome: Proportion of dominated options	
	(1)	(2)
	Increasing menu size from two to three	Reducing menu size from three to two
RDiT estimate	0.007***	-0.053***
	(0.000)	(0.000)
Eff. number of obs.	7,699	5,433
Below-cutoff mean	0.162	0.061

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

However, for a significant proportion of our sample observations (36% of our sample in the first natural experiment, equivalent to 2,793 individuals), the estimated change in the proportion of dominated options within menu does not match the observed choice quality change, as depicted in Table 8. Specifically, for households with two or three members enrolled—expanding menu size does not change the proportion of dominated options within the choice menu. As shown in column (1) in panel A, the point estimate of 0.001 is not statistically significant and it is almost negligible in size compared to the baseline mean of 0.160.

In contrast, column (2) in panel A in Table 8 shows that these enrollees are nonetheless more likely to make choice mistakes after the menu expands. The point estimate for the probability of choosing a strictly dominated option in the same experiment is 0.047, indicating a significant decline in consumer choice quality.[18]

In sum, we show suggestive evidence that the average menu quality driven mechanism alone is not sufficient to explain the results of changing menu size on consumer choice quality that we observe. There is a violation of the pattern for a significant subset of customers.

[18] Appendix Table F.1 presents robustness results without considering riders. For households with two or three members, only one dominated option exists in the two-plan and three-plan menus, meaning the average quality is lower (the proportion of dominated plans is higher) under the two-plan menu. We find that these consumers are still more likely to choose dominated options under the three-plan menu, suggesting that variation in average plan quality is not the driving force.

Table 8: Estimated treatment effects for two- and three-member households

	(1) Increasing menu size from two to three	(2) Reducing menu size from three to two
Panel A. Outcome: Proportion of dominated options		
RDiT estimate	0.001	-0.051***
	(0.001)	(0.000)
Eff. number of obs.	2,793	2,225
Below-cutoff mean	0.160	0.059
Panel B. Outcome: Choosing a dominated plan		
RDiT estimate	0.047*	-0.121***
	(0.027)	(0.029)
Eff. number of obs.	2,793	2,225
Below-cutoff mean	0.143	0.214

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

7. Conclusion

This paper studies how choice set size affects consumer choice quality in private health insurance markets. We leverage a novel dataset and two natural experiments involving a leading health insurance product in China. We find that expanding the plan menu significantly increases the probability of choosing a dominated plan, and shrinking the menu improves choice quality. The estimated treatment effects are symmetric in magnitude. We also provide evidence that these effects are primarily driven by choice overload.

Overall, our results demonstrate that meaningful choice frictions can arise even with relatively small menus in developing country contexts, where consumers are less familiar with health insurance choices and must compare a large number of plan attributes. Presented with a simple menu of only two plans, a significant proportion of our sampled consumers (over 15%) still choose strictly dominated options, although they should have recognized such dominance.

These results highlight the need for careful menu design and well-targeted decision support interventions to improve consumer choice quality in health insurance markets. Based on our findings, one straightforward way to deal with such choice inconsistency would be to reduce the choice set size. However, this approach may have undesirable consequences, as certain consumers may not be able to obtain their preferred options (as recently discussed by Brown and Jeon, 2024). Our further findings on an AI-based decision support intervention suggest a potential pathway to resolve the regulatory trade-off between product diversity and consumer choice quality. The AI-based tool that was provided to agents can bridge the performance gap in consumer decision-making as menu sizes grow larger, thus allowing for a richer variety of plan options without the associated welfare losses from choice overload.

Our findings also shed light on the potential long-term benefits of improved consumer choices. We show that a smaller menu not only results in higher choice quality, but also a higher level of consumer satisfaction, supported by the increased probability of signing up for automatic renewal. This suggests that policies that aim to improve choice quality may also foster long-term consumer retention. For developing countries like China, where private health insurance is an optional entry point into the broader health care sector, interventions that aim to prevent consumer choice mistakes could also contribute to sustainable growth of the industry.

Several limitations point to promising directions for future research. First, our evidence on the choice overload mechanism is indirect. Further analysis would benefit from more granular data on consumers' decision-making processes, including browsing behavior, consultations with agents, and measures of consumer attention. Second, the prevalence of strictly dominated plans raises broader questions about product design and regulation in developing countries. Further research is needed to understand why such plans persist, how they intersect with firms' pricing and marketing strategies, and how regulators should navigate the trade-off between consumer choice quality and the benefits of product variety and innovation.

References

Abaluck, Jason, and Jonathan Gruber. 2011. "Choice Inconsistencies among the Elderly: Evidence from Plan Choice in the Medicare Part D Program." *American Economic Review* 101 (4): 1180–1210.

Abaluck, Jason, and Jonathan Gruber. 2016a. "Choice Inconsistencies among the Elderly: Evidence from Plan Choice in the Medicare Part D Program: Reply." *American Economic Review* 106 (12): 3962–87.

Abaluck, Jason, and Jonathan Gruber. 2016b. "Evolving Choice Inconsistencies in Choice of Prescription Drug Insurance." *American Economic Review* 106 (8): 2145–84.

Abaluck, Jason, and Jonathan Gruber. 2023. "When Less Is More: Improving Choices in Health Insurance Markets." *Review of Economic Studies* 90 (3): 1011–40.

Ang, Geer, Ya Tan, Yingjia Zhai, Fan Zhang, and Qinghua Zhang. 2024. "Housing Wealth, Fertility and Children's Health in China: A Regression Discontinuity Design." *Journal of Health Economics* 97: 102915.

Besedeš, Tibor, Cary Deck, Sudipta Sarangi, and Mikhael Shor. 2015. "Reducing Choice Overload without Reducing Choices." *Review of Economics and Statistics* 97 (4): 793–802.

Bhargava, Saurabh, George Loewenstein, and Justin Sydnor. 2017. "Choose to Lose: Health Plan Choices from a Menu with Dominated Option." *Quarterly Journal of Economics* 132 (3): 1319–72.

Brown, Zach Y., and Jihye Jeon. 2024. "Endogenous Information and Simplifying Insurance Choice." *Econometrica* 92 (3): 881–911.

Bundorf, M. Kate, Maria Polyakova, and Ming Tai-Seale. 2019. "How Do Humans Interact with Algorithms? Experimental Evidence from Health Insurance." NBER Working Paper 25976. National Bureau of Economic Research, Cambridge, MA.

Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. 2024. "Event Studies with a Continuous Treatment." *AEA Papers and Proceedings* 114: 601–05.

Callaway, Brantly, and Pedro HC Sant'Anna. 2021. "Difference-in-differences with multiple time periods." *Journal of econometrics* 225 (2): 200–30.

Calonico, Sebastian, Matlas D. Cattaneo, and Rocio Titiunik. 2014. "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs." *Econometrica* 82 (6): 2295–2326.

Chandra, Amitabh, Benjamin Handel, and Joshua Schwartzstein. 2019. "Behavioral Economics and Health-Care Markets." In *Handbook of Behavioral Economics: Applications and Foundations 1*, Vol. 2, 459–502. North-Holland.

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2023. "Two-way Fixed Effects and Differences-in-Differences Estimators with Several Treatments." *Journal of Econometrics* 236 (2): 105480.

de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2024. "Difference-in-Differences Estimators of Intertemporal Treatment Effects." *Review of Economics and Statistics* Forthcoming.

Ericson, Keith M. Marzilli, Jon Kingsdale, Tim Layton, and Adam Sacarny. 2017. "Nudging Leads Consumers in Colorado to Shop but Not Switch ACA Marketplace Plans." *Health Affairs* 36 (2): 311–19.

Ericson, Keith M. Marzilli, and Amanda Starc. 2016. "How Product Standardization Affects Choice: Evidence from the Massachusetts Health Insurance Exchange." *Journal of Health Economics* 50: 71–85.

Giuntella, Osea, and Fabrizio Mazzonna. 2019. "Sunset Time and the Economic Effects of Social Jetlag: Evidence from US Time Zone Borders." *Journal of Health Economics* 65: 210–26.

Gruber, Jonathan, Benjamin R. Handel, Samuel H. Kina, and Jonathan T. Kolstad. 2020. "Managing Intelligence: Skilled Experts and AI in Markets for Complex Products." NBER Working Paper 27038. National Bureau of Economic Research, Cambridge, MA.

Handel, Benjamin R. 2013. "Adverse Selection and Inertia in Health Insurance Markets: When Nudging Hurts." *American Economic Review* 103 (7): 2643–82.

Handel, Benjamin, Jonathan Kolstad, Thomas Minten, and Johannes Spinnewijn. 2024. "The Socioeconomic Distribution of Choice Quality: Evidence from Health Insurance in the Netherlands." *American Economic Review: Insights* 6 (3): 395–412.

Hausman, Catherine, and David S. Rapson. 2018. "Regression discontinuity in time: Considerations for empirical applications." *Annual Review of Resource Economics* 10 (1): 533–52.

Iyengar, Sheena S., and Mark R. Lepper. 2000. "When Choice Is Demotivating: Can One Desire Too Much of a Good Thing?" *Journal of Personality and Social Psychology* 79 (6): 995.

Ketcham, Jonathan D., Nicolai V. Kuminoff, and Christopher A. Powers. 2016. "Choice Inconsistencies among the Elderly: Evidence from Plan

Choice in the Medicare Part D Program: Comment." *American Economic Review* 106 (12): 3932–61.

Ketcham, Jonathan D., Claudio Lucarelli, Eugenio J. Miravete, and M. Christopher Roebuck. 2012. "Sinking, Swimming, or Learning to Swim in Medicare Part D." *American Economic Review* 102 (6): 2639–73.

Ketcham, Jonathan D., Claudio Lucarelli, and Christopher A. Powers. 2015. "Paying Attention or Paying Too Much in Medicare Part D." *American Economic Review* 105 (1): 204–33.

Kling, Jeffrey R., Sendhil Mullainathan, Eldar Shafir, Lee C. Vermeulen, and Marian V. Wrobel. 2012. "Comparison Friction: Experimental Evidence from Medicare Drug Plans." *Quarterly Journal of Economics* 127 (1): 199–235.

Lee, Ajin, and Boris Vabson. 2024. "The Value of Improving Insurance Quality: Evidence from Long-run Medicaid Attrition." *Journal of Health Economics* 94: 102865.

Li, Honglin, Chenyuan Liu, and Justin Sydnor. 2025. "Is Choice Overload a Problem for Health-Plan Choice? Evidence from the ACA Marketplace." Working paper.

Liu, Chenyuan, and Justin Sydnor. 2022. "Dominated Options in Health Insurance Plans." *American Economic Journal: Economic Policy* 14 (1): 277–300.

Liu, Hong, Song Gao, and John A. Rizzo. 2011. "The Expansion of Public Health Insurance and the Demand for Private Health Insurance in Rural China." *China Economic Review* 22 (1): 28–41.

McCrary, Justin. 2008. "Manipulation of the Running Variable in the Regression Discontinuity Design: A Density Test." *Journal of Econometrics* 142 (2): 698–714.

Samek, Anya, and Justin Sydnor. 2025. "The Impact of Consequence Information on Insurance Choice." *Review of Economics and Statistics* Forthcoming.

Scheibehenne, Benjamin, Rainer Greifeneder, and Peter M. Todd. 2010. "Can There Ever Be Too Many Options? A Meta-analytic Review of Choice Overload." *Journal of Consumer Research* 37 (3): 409–25.

Zhu, Minglai, Youji Lyu, Benke Wang, Jingjing Ou, Rui Zhu, Shanshan Wang, Zihan Diao, et al. 2024. "Blue Paper on the Development of China's Commercial Medical Insurance." Accessed December 13, 2025,

https://mp.weixin.qq.com/s/DButxC1wabe1fejwWDrVBg?scene=25&sessionid=#wechat_redirect (in Chinese).

Appendix (For Online Publication Only)

A. Plan details and strictly dominated options

This section provides details on the sampled plans and the strictly dominated options in each menu.

A.1 Coverage details

Our analysis includes four plans. Plan A, Plan B, and Plan D were launched at the beginning of 2019, 2020, and 2021, respectively. Plan C is a family plan that was launched in mid-2020. These four plans cover inpatient medical expenses and expenses associated with critical illness treatment which exceed the cap of public basic medical insurance coverage. The following are the differences in coverage details across plans.

Benchmark for comparison. Plan A includes a 30-day waiting period and has a high deductible (10,000 yuan or roughly $1,450) for inpatient expenses and no deductible applies to expenses related to critical illness. The plan has a 100% compensation ratio above the deductible, a 300 million yuan (or roughly $435,000) upper limit for inpatient expenses, and another 300 million yuan coverage for critical illness. Outpatient services associated with inpatient admissions—specifically, those provided in the seven days preceding and 30 days following hospitalization—are deemed part of the same treatment episode and therefore covered by the plan.

Differences in coverage details. The four products offer identical insurance benefits, except for: (i) treatments for critical illness: compared to Plan A, the other plans (Plan B, Plan C, and Plan D) expand the list of covered critical illnesses by including 121 rare diseases defined by the government (e.g., albinism)[19]; (ii) shared deductible with family members: compared to enrollees of other plans (Plan A, Plan B, and Plan D), enrollees in Plan C have a 10,000 yuan deductible associated with inpatient expenses that can be shared with their family members; (iii) outpatient services: Plan D covers outpatient services provided in the 30 days preceding and 30 days following a claimed inpatient visit. In comparison, its predecessors (Plan A, Plan B, and Plan C) only cover outpatient services within seven days preceding and 30 days following an inpatient visit.

[19] See the official website of the National Health Commission of the People's Republic of China for details: https://www.gov.cn/zhengce/zhengceku/2018-12/31/content_5435167.htm.

A.2 Strictly dominated plans without consideration of the rider choices

The existence of strictly dominated plans in a menu depends on the enrollee's age, claim history, and household size.

Age and baseline premium. For a given insurance plan, the baseline premium is mainly determined by an enrollee's age at renewal.[20]

Claim history, household size, and premium discount. The gross premium paid by an enrollee at renewal equals the baseline premium minus the premium discount. Plan A's enrollees are not eligible for any discounts, but enrollees in Plan B, Plan C, and Plan D can benefit from a premium discount if they have no claim history ("zero-claim enrollees"). The discount is fixed at 5% for all zero-claim enrollees in Plan B. Under Plan C and Plan D, the discount rates are 5%, 10%, and 15% for zero-claim enrollees with fewer than three, exactly three, and more than three enrolled zero-claim family members, respectively.

A.3 Strictly dominated plan-by-rider options

Some health insurance plans provide rider options for specific medical catalogs and services. In the baseline regressions, we take the choice of riders into consideration to measure consumer choice quality by exploring all available plan-by-rider combinations. The total number of available options (plan-by-rider combinations) is as follows: $Total\ options = \sum_m 2^{n_m}$, where m indexes insurance plans within a menu and n_m is the total number of available riders under plan m.

Table A.1 reports the total number of available options in our setting. Although a plan's menu of riders can change over time, no such change took place within the sampled time windows of the two natural experiments (within a 60-day bandwidth around the cutoff dates).

[20] Public basic medical insurance enrollment status was also a determinant of the baseline premium, but it did not change the ordinal ranking of gross premiums (the baseline premium net of the premium discount) and thus did not affect plan dominance. In addition, according to the insurer, approximately 99% of consumers of the sampled product were enrolled in the public basic medical insurance. Thus, we do not take this factor into account in our analysis.

Table A.1: Total number of available plan-by-rider options

Menu	Plan	Household size = 1		Household size > 1	
		Age <= 17	Age > 17	Age <= 17	Age > 17
Panel A. Natural experiment 1: Increasing choice set size from two to three					
Menu 1	Plan A	2^6	2^5	2^7	2^6
	Plan B	2^6	2^6	2^7	2^7
Menu 2	Plan A	2^6	2^5	2^7	2^6
	Plan B	2^6	2^6	2^7	2^7
	Plan C	2^0	2^0	2^0	2^0
Panel B. Natural experiment 2: Reducing choice set size from three to two					
Menu 2	Plan A	2^5	2^4	2^6	2^5
	Plan B	2^8	2^8	2^9	2^9
	Plan C	2^0	2^0	2^0	2^0
Menu 3	Plan A	2^5	2^4	2^6	2^5
	Plan D	2^{11}	2^{11}	2^{12}	2^{12}

Notes: Panels A and B report the choice sets of enrollees who renewed within the time windows around the two natural experiments—within a 60-day bandwidth around the cutoff dates. The choice set size differs by age group due to a plan-specific rider for additional accident coverage, which is only available to Plan A enrollees under age 18. The choice set size differs by household size because the shared-deductible rider (with one's enrolled family members) under Plan A, Plan B, and Plan D is unavailable for single enrollees. Plan C does not allow for the purchase of riders.

Sample restrictions to ensure the existence of strictly dominated plans. A plan-by-rider combination strictly dominates another in our setting if it (i) offers the same or better insurance protection (e.g., a broader network), and (ii) charges the same or lower premium, with at least one of the two conditions being strictly favorable to the consumer.

We exclude two groups of enrollees from our analysis who did not face any strictly dominated plans under all three menus as shown in Figure 1. The first group consists of enrollees who had claim histories at the time of renewal and were thus not eligible for premium discounts under Plan B, Plan C, or Plan D.[21] Premium discounts are necessary for these new plans to dominate Plan A. We also remove

[21] Since we do not directly observe whether a sampled insured individual had a claims history at the time of renewal in the natural experiments, we exclude individuals whose associated insurance agents had "claimed clients." The insurance company defines an agent as having "claimed clients" if at least one of the agent's historical clients had a claims record.

enrollees age between 16 and 20 and those age over 60. For these two age groups, the lower-coverage plan (Plan A) charged a lower gross premium under the two-plan menus (Menu 1 and Menu 3). Hence, strict plan dominance does not exist.

Table A.2 reports the total number of strictly dominated plan-by-rider options for enrollees aged 0-15 and 21-60 without a claim history at renewal (the main study sample).

Table A.2: Total number of strictly dominated plan-by-rider options

Menu	Plan	Household size = 1	Household size = 2	Household size = 3	Household size > 3
Menu 1	Plan A	2^4	2^5	2^5	2^5
	Plan B	0	0	0	0
Menu 2	Plan A	2^4	2^5	2^5	2^5
	Plan B	0	0	1	2
	Plan C	1	0	0	0
Menu 3	Plan A	2^4	2^5	2^5	2^5
	Plan D	0	0	0	0

B. Identification validity

B.1 Density tests

We test for discontinuities in the density of the running variable at the cutoffs, following McCrary (2008). The tests are based on the main study sample of individuals within a 60-day bandwidth around the cutoffs. Panels A and B in Figure B.1 present graphical evidence for the two natural experiments (increasing and reducing choice set sizes), respectively. The corresponding estimates (standard errors) for the discontinuities at the cutoffs are 0.001 (0.047) and 0.075 (0.046), respectively.

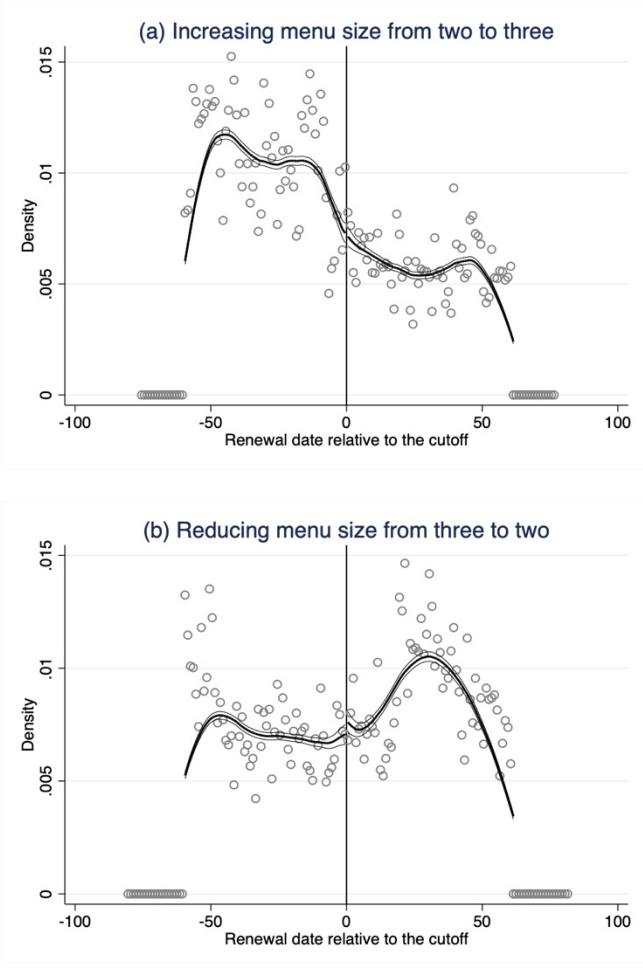


Figure. B.1: Density tests

B.2 Covariate balance tests

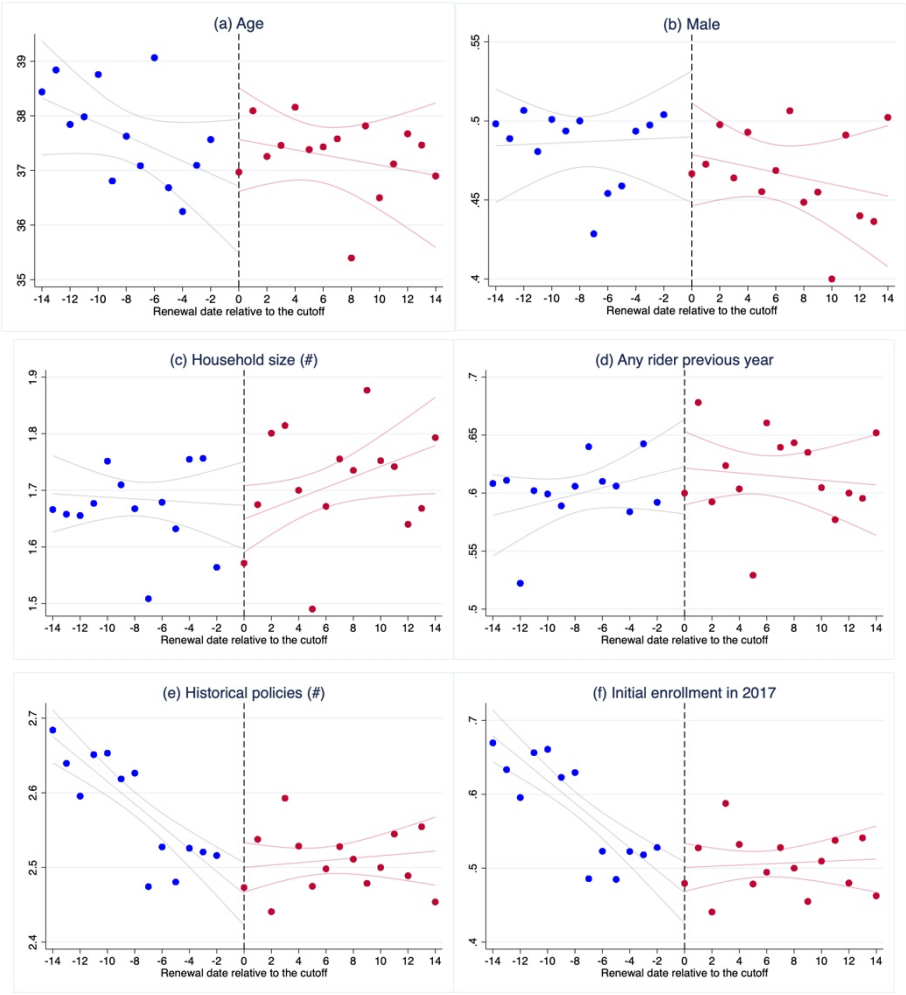


Figure. B.2: Covariate balance (increasing menu size from two to three)

Notes: Each figure plots the means of the outcomes at each standardized renewal date bin, together with fitted lines and 95% confidence intervals. The fitted lines are produced using regressions with a triangular kernel. The first and second choice menus are indicated in blue and red, respectively.

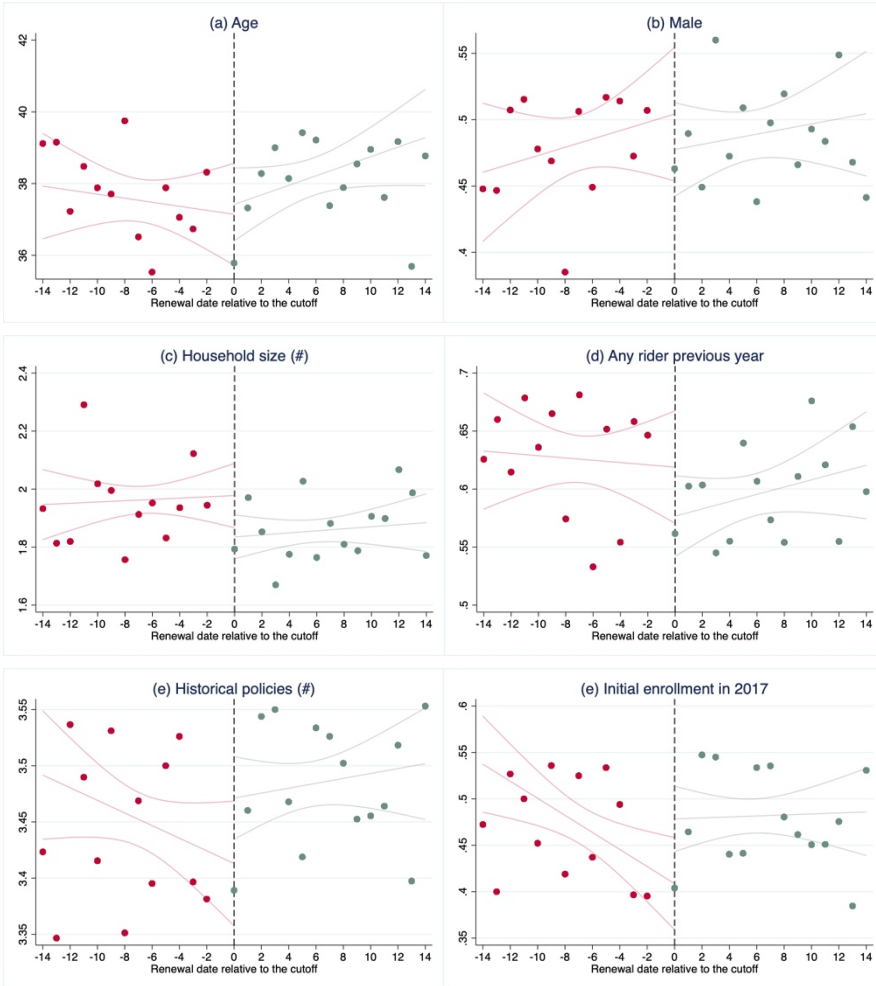


Figure. B.3: Covariate balance (reducing menu size from three to two)

Notes: Each figure plots the means of the outcomes at each standardized renewal date bin, together with fitted lines and 95% confidence intervals. The fitted lines are produced using regressions with a triangular kernel. The second and third choice menus are indicated in red and green, respectively.

C. Testing for changes in the renewal rate

This section tests whether there are changes in the renewal rate at the cutoff dates. We employ a complementary data set that was acquired from the same data source, which contains information on the Plan A policies that were purchased a year before the natural experiments took place. These policies were scheduled for renewal around the cutoff dates. We implement the same sample restrictions as in our baseline empirical analysis to ensure the existence of strictly dominated options within each menu and to rule out potential confounders, to ensure cleaner estimates (as reported in section 3).

We utilize the constructed sample to test whether the renewal rate changes significantly at the cutoffs. We focus on the expiration date because the enrollees usually renewed around this date.[22] We specify the running variable as the individual's policy expiration date relative to the cutoff dates of the menu changes. To ensure comparability, we restrict the sample to individuals whose policy expiration dates were within the 60-day bandwidth before and after the cutoff dates.

For estimation, we use the baseline RDiT strategy presented in section 4 and replace the outcome variable with a dummy indicator for plan renewal. Optimal bandwidth selection method of Calonico et al. (2014) yields optimal bandwidths of around 14 days. Thus, we set a uniform bandwidth of 14.

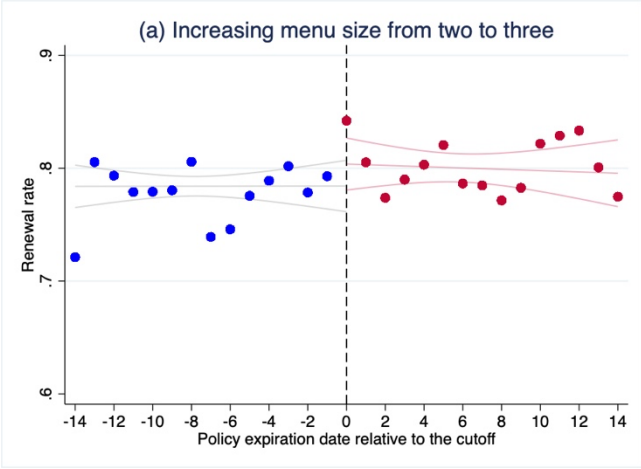
Table C.1 reports the estimates. The results indicate that the renewal rate barely changes at either cutoff. The estimated changes are 0.019 and 0.004 for the first and second natural experiments, respectively. Both estimates are insignificant and of small magnitude compared to the baseline means below the cutoffs. The graphical evidence presented in Figure C.1 shows consistent findings, with no obvious discontinuities at either cutoff.

[22] In our main study sample of renewed enrollees in the two natural experiments, around 55% of them renewed within 10 days around the expiration date.

Table C.1: Testing for changes in the renewal rate

	Outcome: Choosing to renew	
	(1)	(2)
	Increasing menu size from two to three	Reducing menu size from three to two
RDiT estimate	0.019	0.004
	(0.017)	(0.027)
Eff. number of obs.	14,234	5,415
Below-cutoff mean	0.785	0.788

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.



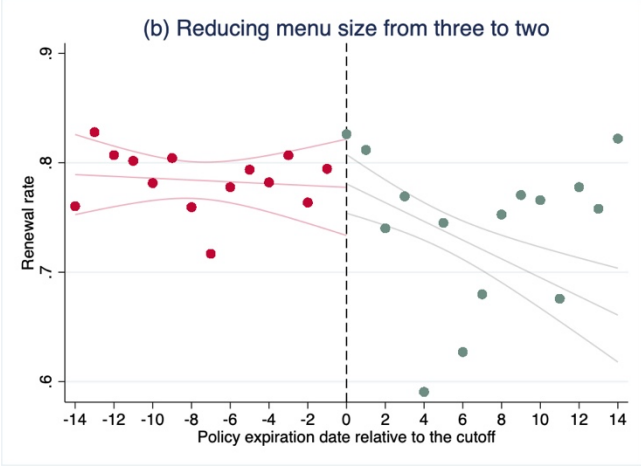


Figure. C.1: Insurance renewal rates around the cutoffs

Notes: Each figure plots the renewal rate at each standardized policy expiration date bin, together with fitted lines and 95% confidence intervals. The fitted lines are produced using regressions with a triangular kernel. The first, second, and third choice menus are indicated in blue, red, and green, respectively.

D. Specification checks

Table D.1: Sensitivity of main results

	Outcome: Choosing a strictly dominated plan					
	(1) Baseline	(2) Donut hole (radius = 1)	(3) Quadratic polynomial	(4) Uniform kernel	(5) Optimal bandwidth	(6) Add covariates
Panel A. Increasing menu size from two to three						
RDiT estimate	0.084***	0.081***	0.087**	0.097***	0.087***	0.084***
	(0.023)	(0.027)	(0.034)	(0.020)	(0.020)	(0.024)
Eff. number of obs.	7,699	7,092	7,699	8,480	10,491	7,699
Bandwidth	14	14	14	14	17.892	14
Kernel	Triangular	Triangular	Triangular	Uniform	Triangular	Triangular
Polynomial order	1	1	2	1	1	1
Covariates	N	N	N	N	N	Y
Below-cutoff mean	0.161					
Panel B. Reducing menu size from three to two						
RDiT estimate	-0.110***	-0.110***	-0.159***	-0.086***	-0.117***	-0.114***
	(0.031)	(0.032)	(0.046)	(0.028)	(0.032)	(0.028)
Eff. number of obs.	5,433	4,991	5,433	5,775	4,758	5,433
Bandwidth	14	14	14	14	11.368	14
Kernel	Triangular	Triangular	Triangular	Uniform	Triangular	Triangular
Polynomial order	1	1	2	1	1	1
Covariates	N	N	N	N	N	Y
Below-cutoff mean	0.225					

Notes: Panels A and B report the results from the natural experiments of increasing and reducing choice set size, respectively. Column (1) reports the baseline estimates. Columns (2) to (6) further report the results of implementing a donut-hole RDiT (excluding the nearest observations within a one-day radius around the cutoff), using a quadratic polynomial, utilizing a uniform kernel, employing the optimal bandwidth as suggested by Calonico et al. (2014) for each regression, and adding covariates. "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D.2: Sensitivity of placebo test results

	Outcome: Choosing a strictly dominated plan					
	(1) Baseline	(2) Donut hole (radius = 1)	(3) Quadratic polynomial	(4) Uniform kernel	(5) Optimal bandwidth	(6) Add covariates
RDiT estimate	0.006	0.007	0.010	0.004	0.009	0.006
	(0.013)	(0.014)	(0.019)	(0.013)	(0.015)	(0.013)
Eff. number of obs.	10,362	7,807	10,362	10,732	8,994	10,362
Bandwidth	14	14	14	14	10.351	14
Kernel	Triangular	Triangular	Triangular	Uniform	Triangular	Triangular
Polynomial order	1	1	2	1	1	1
Covariates	N	N	N	N	N	Y
Below-cutoff mean	0.071					

Notes: Column (1) reports the baseline estimates. Columns (2) to (6) further report the results of implementing a donut-hole RDiT (excluding the nearest observations within a one-day radius around the cutoff), using a quadratic polynomial, utilizing a uniform kernel, employing the optimal bandwidth as suggested by Calonico et al. (2014), and adding covariates (age, gender and household size). "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who enrolled within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

E. Evidence for the choice overload mechanism

Table E.1: Heterogeneous treatment effects by enrollment experience

	Outcome: Choosing a dominated option	
	(1) Increasing menu size from two to three	(2) Reducing menu size from three to two
Panel A. More-experienced consumers		
RDiT estimate	0.030	-0.028
	(0.029)	(0.044)
Eff. number of obs.	4,217	3,482
Below-cutoff mean	0.151	0.176
Panel B. Less experienced consumers		
RDiT estimate	0.141***	-0.171***
	(0.030)	(0.027)
Eff. number of obs.	2,520	2,913
Below-cutoff mean	0.200	0.246

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

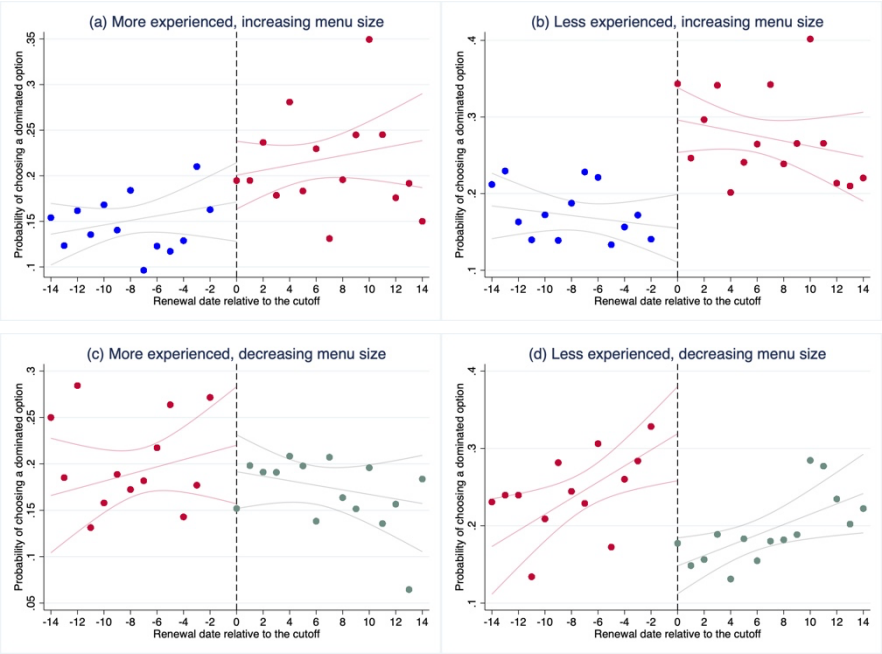


Figure E.1: Heterogeneous treatment effects by enrollment experience

Notes: Each figure plots the probability of choosing a dominated plan at renewal at each standardized renewal-date bin, together with fitted lines and 95% confidence intervals. The fitted lines are produced using regressions with a triangular kernel. The first, second, and third choice menus are indicated in blue, red, and green, respectively.

Table E.2: Effect of decision support on choice quality by plan menu

	Outcome: Choosing a dominated option		
	(1) Baseline	(2) Not-yet-treated group as control	(3) TWFE
Panel A. Choosing under Menu 2 (three-plan menu)			
DID estimate	-0.221**	-0.252**	-0.223***
	(0.109)	(0.112)	(0.069)
Obs.	47,046	2,084	47,046
Pre-treatment mean		0.294	
Panel B. Choosing under Menu 3 (two-plan menu)			
DID estimate	0.038	0.037	0.038
	(0.062)	(0.088)	(0.063)
Obs.	15,079	614	15,079
Pre-treatment mean		0.112	

Notes: Panels A and B report results based on the subsamples of individuals choosing under Menu 2 and Menu 3, respectively. Column (1) reports the baseline estimate. Columns (2) and (3) report results using the not-yet-treated group as control and employing a two-way fixed effects (TWFE) estimator. "Pre-treatment mean" reports the unconditional sample mean of the outcome for treatment-group policyholders during the pre-treatment period.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

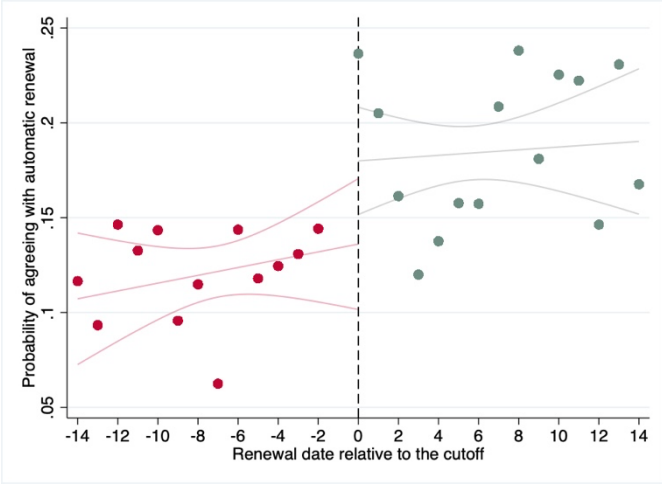


Figure E.2: Reducing menu size and consumer satisfaction

Notes: This figure plots the probability of agreeing with automatic renewal of the currently chosen options at each standardized renewal-date bin, together with fitted lines and 95% confidence intervals. The fitted lines are produced using regressions with a triangular kernel. The second and third choice menus are indicated in red and green, respectively.

F. Evidence against the "average plan quality change" mechanism

Table F.1: Treatment effects for two- and three-member households without considering the choice of riders

	(1) Increasing menu size from two to three	(2) Reducing menu size from three to two
Panel A. Outcome: Proportion of dominated options		
RDiT estimate	-0.167***	0.167***
	(0.000)	(0.000)
Eff. number of obs.	2,793	2,225
Below-cutoff mean	0.500	0.333
Panel B. Outcome: Choosing a dominated plan		
RDiT estimate	0.039	-0.119***
	(0.029)	(0.032)
Eff. number of obs.	2,793	2,225
Below-cutoff mean	0.146	0.210

Notes: "Below-cutoff mean" is the unconditional sample mean of the outcome for policyholders who renewed within a 14-day bandwidth below the cutoff.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.